

- No equipment may be borrowed during the exam
- All necessary work must be shown. Marks may be deducted for careless or badly arranged work.

Question 1 (Start a new page)

- (a) Write $1 + \frac{1+i}{1-i}$ in the form $a + ib$ where a and b are real.
- (b) Find the two square roots of $-7 + 24i$
- (c) Shade on an Argand diagram the region $|z| < |z - 2 + i|$
- (d) Describe, in geometric terms, the locus (in the Argand Plane) represented by $|z - 5i| = 3$
- (e) Find the complex number z with the least positive argument satisfying the condition $|z - 5i| \leq 3$

Question 2 (Start a new page)

- (a) Define the modulus $|z|$ of a complex number z and show that every non-zero complex number z can be written in the form $r(\cos \theta + i \sin \theta)$ where $r = |z|$.
- (b) Express $\frac{1}{1+i}$ in the form described in (a)
- (c) If $z = r(\cos \theta + i \sin \theta)$, show that $\frac{z}{z^2 + r^2}$ is purely real and give its value
- (d) The four complex numbers z_1, z_2, z_3, z_4 are represented on the complex (Argand) plane by the points A, B, C, D respectively. If $z_1 - z_2 + z_3 - z_4 = 0$ and $z_1 - i z_2 - z_3 + i z_4 = 0$, determine the possible shape (s) for the quadrilateral ABCD

Question 3 (Start a new page)

- (a)(i) Find the cube roots of unity expressing them in mod-arg form. Show the roots on an argand diagram.
- (ii) If w is one of the complex roots, prove the other root is w^2 , and show that $1 + w + w^2 = 0$
- (iii) Prove that $(1 + w - w^2)^3 - (1 - w + w^2)^3 = 0$
- (iv) Prove that if n is a positive integer, then $1 + w^n + w^{2n} = 3$ or 0 , depending on whether n is or is not a multiple of 3 .
- (b) Find the greatest value of the moduli of complex numbers z satisfying the equation

$$\left| z - \frac{4}{z} \right| = 2$$

QUESTION 4 (Start a new page)

(a) P , Q and R represent the complex numbers z_1, z_2, z_3 respectively, where $z_1 z_2 = z_3^2$. Show geometrically that OR bisects the angle POQ

(b) A variable point P on the Argand diagram represents the complex number z . α and β are fixed complex numbers. Describe and illustrate with sketches the geometrical relations between the point P and :

- (i) p_1 representing $z + \alpha$
- (ii) p_2 representing βz

(c) A sequence of points $z_1, z_2, \dots$ is defined from an arbitrary point z_0 by the transformations

$$z_n = c \bar{z}_{n-1} + c - 1$$

where $c = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$.

Determine whether the sequence consists of repetitions of a finite set of points, and if so how many points there are in the set

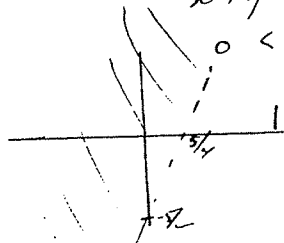
THIS IS THE END OF THE TEST

$$\begin{aligned}
 a) 1 + \frac{1-i}{1-i} &= \frac{1-i}{1-i} \\
 &= \frac{2}{1-i} \\
 &= \frac{2}{1-i} \cdot \frac{1+i}{1+i} \\
 &= \frac{2+2i}{2} \\
 &= 1+i
 \end{aligned}$$

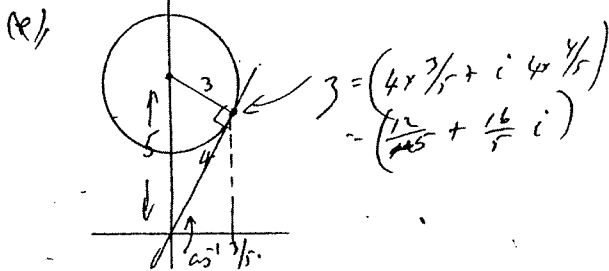
$$\begin{aligned}
 (b) -7+24i &= a^2 - b^2 + i2ab \\
 \therefore a^2 - b^2 &= -7 \\
 ab &= 12 \\
 \therefore a^2 - \frac{144}{a^2} &= -7 \\
 a^4 + 7a^2 - 144 &= 0 \\
 (a^2 + 16)(a^2 - 9) &= 0 \\
 a &= \pm 3
 \end{aligned}$$

$$\therefore \text{roots } \pm(3+4i)$$

$$\begin{aligned}
 (c) z = x+iy \quad |z| < |3-2+i| \\
 x^2 + y^2 < (x-2)^2 + (y+1)^2 \\
 0 < -4x + 2y + 5
 \end{aligned}$$



$$(d) \text{ Circle centre } (0, 5) \text{ radius } 2$$

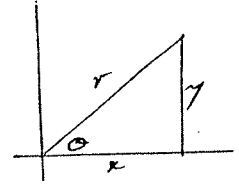


$$4.7.4.26$$

3

$$4.2 \quad (a) \text{ given } z = x+iy \text{ then } |z| = \sqrt{x^2+y^2}$$

$$z = x+iy = r(\cos \theta + i \sin \theta)$$

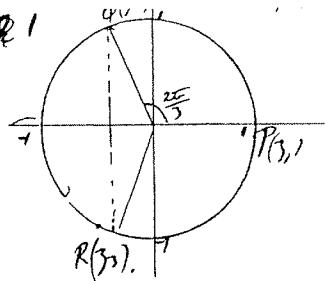


$$\begin{aligned}
 (b) \frac{1}{1+i} \\
 &= \frac{1}{1+i} \cdot \frac{1-i}{1-i} \\
 &= \frac{1-i}{2} \\
 &= \frac{1}{2} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})
 \end{aligned}$$

$$\begin{aligned}
 (c) \frac{r(\cos \theta + i \sin \theta)}{r^2(\cos 2\theta + i \sin 2\theta) + r^2} \\
 &= \frac{1}{r} \frac{\cos \theta + i \sin \theta}{(\cos 2\theta + i \sin 2\theta) + 1} \\
 &= \frac{1}{r} \frac{\cos \theta + i \sin \theta}{2\cos^2 \theta + i 2\sin \theta \cos \theta} \\
 &= \frac{1}{r} \frac{\cos \theta + i \sin \theta}{2\cos \theta} \text{ points in quadrant}
 \end{aligned}$$

$$\begin{aligned}
 (d) \text{ If } z_1 - z_2 = z_4 - z_3 \text{ then } AB \text{ \& } DC \text{ are equal \& parallel.} \\
 \text{If } z_1 - z_3 = i(z_2 - z_4) \text{ then } AC \text{ \& } BD \text{ are equal \& perp.} \\
 \therefore ABCD \text{ is a square.}
 \end{aligned}$$

Q3 a(i) $1, \cos \frac{2\pi}{3}, \cos \frac{4\pi}{3}$
 let roots be z_1, z_2, z_3 represented
 by P, Q, R .



(ii) z_1 is real

If $z_1 = \omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

then $z_3 = \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right)$
 $= \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^2$

+ $i \sin \frac{2\pi}{3}$

& Since $1, \omega, \omega^2$ are the roots of $z^3 - 1 = 0$
 then the sum of the roots = 0.

$\therefore 1 + \omega + \omega^2 = 0$

(iii) $(1 + \omega - \omega^2)^3 - (1 - \omega + \omega^2)^3$
 $= (-2\omega^2)^3 - (-2\omega)^3$
 $= -8\omega^6 + 8\omega^3$
 $= 8\omega^3(1 - \omega^3)$

$\omega^3 = 1$ since ω is a root of $z^3 = 1$

(iv) let n be of form $3K, 3K+1, 3K+2$.

If $n = 3K$
 $1 + \omega^n + \omega^{2n}$
 $= 1 + \omega^{3K} + \omega^{6K}$
 $= 1 + (\omega^3)^K + (\omega^3)^{2K}$
 $= 1 + 1 + 1$
 $= 3$

If $n = 3K+1$
 $1 + \omega^n + \omega^{2n}$
 $= 1 + \omega^{3K+1} + \omega^{6K+2}$
 $= 1 + \omega \cdot \omega^{3K} + \omega^2 \cdot \omega^{6K}$
 $= 1 + \omega + \omega^2$
 $= 0$

If $n = 3K+2$
 $1 + \omega^n + \omega^{2n}$
 $= 1 + \omega^{3K+2} + \omega^{6K+4}$
 $= 1 + \omega^2 \cdot \omega^{3K} + \omega^4 \cdot \omega^{6K}$
 $= 1 + \omega^2 + \omega$
 $= 0$

3b. Consider a $\triangle$ with side lengths $|z|, |\frac{z}{3}|, |z - \frac{z}{3}|$

Consider triangle inequality where $|z|$ is the largest side

$= |z| + |\frac{z}{3}| \geq |z - \frac{z}{3}|$

$|\frac{z}{3}| + 2 \geq |z|$

$|z|^2 - 2|z| \leq 4$

Complete the square

$|z| \leq \sqrt{5} + 1$

$\therefore$ greatest value is $\sqrt{5} + 1$

Q4 (a) If $z_1, z_2 = z_3$

then $\arg z_1 + \arg z_2 = 2 \arg z_3$

Let $\alpha = \arg z_1$

$\beta = \arg z_2$

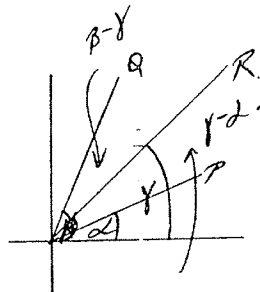
$\gamma = \arg z_3$

$\therefore$ We are given that $\alpha + \beta = 2\gamma$

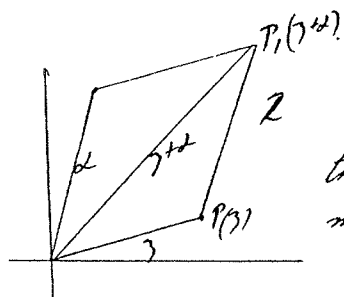
$\therefore \alpha - \gamma = \gamma - \beta$

$\therefore$ OR $\arg(100)$

4



b) (i)

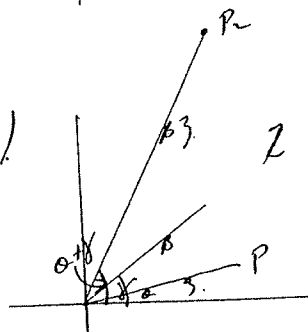


If $\alpha = a + ib$

2

then P_1 is obtained from P by moving a units parallel to real & b units parallel to y axis.

(ii)



P_2 is obtained from P by a rotation of equal to $\arg B$ & an enlargement by a factor of $|b|$.

2

(c)

$$z_1 = c \bar{z}_0 + c - 1$$

$$\bar{z}_1 = \bar{c} \bar{z}_0 + \bar{c} - 1$$

$$z_2 = c \bar{z}_1 + c - 1$$

$$= c(\bar{c} \bar{z}_0 + \bar{c} - 1) + c - 1$$

$$= z_0$$

3

Sequence has repetitions z_0 & z_1 .

3b

Consider a Δ with side lengths $|z|$, $|\frac{4}{3}|$ & $|3 - \frac{4}{3}|$

Consider triangle inequality where $|z|$ is the large side

$$\therefore |\frac{4}{3}| + |3 - \frac{4}{3}| \geq |z|$$

$$|\frac{4}{3}| + 2 \geq |z|$$

$$|z|^2 - 2|z| \leq 4$$

Complete the square

$$|z| \leq \sqrt{5} + 1$$

$\therefore$ greatest value is $\sqrt{5} + 1$

4

14 (a) If $z_1, z_2 = z_3$

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Let $\alpha = \arg z_1$

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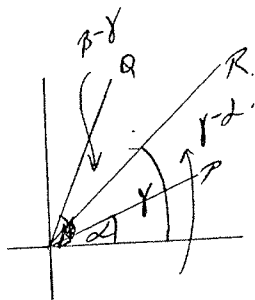
$\gamma = \arg z_3$

$\therefore$ We are given that $\alpha + \beta = 2\gamma$

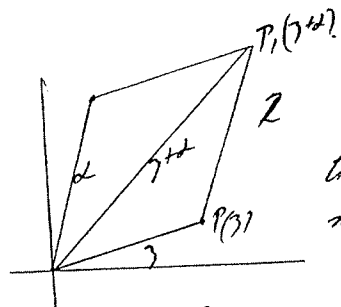
$$\therefore \alpha - \gamma = \gamma - \beta$$

$\therefore$ OR $\arg(90^\circ)$

4

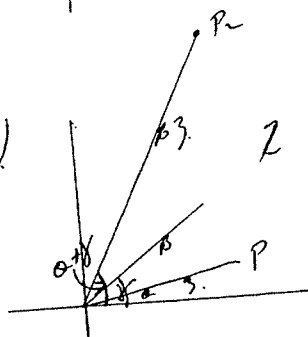


b. (i)



If $\alpha = a+ib$ then P_1 is obtained from P by moving a units parallel to real axis & b units parallel to y axis.

(ii)



P_2 is obtained from P by a rotation of equal to $\arg B$ & an enlargement by a factor of $|b|$.

(c)

$$z_1 = c \bar{z}_0 + c - 1$$

$$z_2 = c \bar{z}_1 + c - 1$$

$$= c(\bar{c} z_0 + \bar{c} - 1) + c - 1$$

$$= z_0$$

Sequence has repetitions z_0 & z_1 .

$$\bar{z}_1 = \bar{c} z_0 + \bar{c} - 1$$

3